MIL-C-47113B(MI)
28 November 1986
SUPERSEDING
MIL-C-47113A(MI)
16 April 1982

MILITARY SPECIFICATION

COMPOUND, HEAT SINK, SILICONE AND/OR NON-SILICONE

This specification is approved for use by the U.S. Army Missile Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

*1.1 Scope. This specification covers a silicone and/or non-silicone heat sink compound which is applied to the base and mounting studs of transistors and diodes to provide a positive heat sink seal (see 6.1).

*1.2 Classification. Heat sink compound covered by this specification shall be of the following types as specified (see 6.2).

Type I - Silicone
Type II - Non-silicone

*1.3 Military part number. The military part number shall consist of the letter B, the basic number of this specification, and dash numbers denoting type and container size.

Example: B47113 - I OZ

- B47113 — Basic spec no.
- I — Type (see 1.2)
- OZ — Container size (Table I and 5.1.1)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5276, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) at the end of this document or by letter.

AMSC N/A FSC 6850
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*TABLE I. Container size

Container	Dash identification character
OUNCE	OZ
PINT	PT
QUART	QT
GALLON	GL
POUND	LB

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

VV-D-1078 Damping Fluid, Silicone Base (Dimethyl Polysiloxane)

STANDARDS

FEDERAL

FED-STD-791 Lubricant, Liquid Fuel and Related Products, Methods of Testing

MILITARY

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

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*2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 70 Specific Gravity of Semi-Solid Bituminous Materials, Test Method for

ASTM D 217 Cone Penetration of Lubricating Grease

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

*2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

*3.1 First article. When specified, (see 6.2), the contractor shall furnish first article samples for first article inspection in accordance with 4.3. The first article samples shall be produced by the same methods, processes and techniques to be employed in subsequent production of the contract quantity.

3.2 Material.

3.2.1 Composition.

*3.2.1.1 Composition. The compound shall be composed of the following materials:

a. Type I:

Base Fluid - Dimethyl polysiloxane in accordance with VV-D-1078, viscosity grade as appropriate to meet the requirements of paragraphs 3.3 and 3.4.

Filler - Zinc Oxide 99.0 + % pure - fine particle size

b. Type II:

Base Fluid - Polyol Ester
Other Materials - Zinc Oxide, Aluminum Silicate

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3.2.1.2 Condition in container. The material in containers shall be free from foreign matter that would adversely affect its intended purpose.

3.2.2 Specific gravity. The material shall have a specific gravity of not less than 2.3.

3.2.3 Thermal conductivity. The compound shall exhibit a thermal conductivity (K factor) of not less than 0.0005 as defined by the expression below:

K shall be in units of cal/sec - cm^2.°C.

3.2.4 Color. The material shall be opaque white.

3.3 Performance characteristics.

3.3.1 Penetration. The penetration shall measure not less than 250 and not more than 320.

3.4 Environmental conditions.

3.4.1 Bleed. Bleed shall not exceed 1.0 percent.

3.4.2 Evaporation. Evaporation shall not exceed 2.0 percent.

3.5 Workmanship. The compound, as packaged in containers, shall be a smooth, homogeneous mixture, free of lumps and foreign matter when examined visually.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

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*4.3 First article inspection. When specified in the contractual document (see 6.2), a sample consisting of 4, 6 or 8 fluid ounces in sealed tubes or cans, shall be subjected to first article inspection. If the first article sample does not meet the requirements of this specification it shall be rejected. Subsequent units shall not be considered for acceptance until Government approval of the first article sample has been obtained. Units subjected to first article inspection shall have successfully passed acceptance inspection. Testing of the first article sample, to determine compliance with the characteristics listed in Table II, shall be conducted in accordance with the corresponding test paragraphs.

*Table II

First Article Inspection		
Characteristics	Requirement Source	Test paragraph
Composition and condition in container	3.2.1	4.6.3.1
Specific gravity	3.2.2	4.6.3.2
Thermal conductivity	3.2.3	4.6.3.3
Color	3.2.4	4.6.3.4
Penetration	3.3.1	4.6.3.5
Bleed	3.4.1	4.6.3.6
Evaporation	3.4.2	4.6.3.7
Workmanship	3.5	4.6.4

4.4 Quality conformance inspection.

4.4.1 Sampling for acceptance inspection. Sampling for acceptance inspection shall be conducted in accordance with MIL-STD-105, at an acceptable quality level (AQL) of 4.0 percent defective.

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4.4.2 Lot formation. A lot shall consist of all the material manufactured in one continuous operation by the same process, by the same manufacturer in accordance with this specification and submitted for inspection at one time.

*4.4.3 Acceptance inspection. Acceptance inspection of the sample specified in 4.4.1, to determine compliance with the characteristics in Table III, shall be conducted in accordance with the corresponding test and inspection paragraphs.

Table III

Acceptance Inspection		
Characteristics	Requirement Source	Test paragraph
Composition and condition in container	3.2.1	4.6.3.1
Specific gravity	3.2.2	4.6.3.2
Workmanship	3.5	4.6.4
Packaging, packing, and marking	5.	4.7

4.5 Inspection equipment. The inspection equipment for conducting examination and tests shall be as specified in the applicable test methods and procedures paragraphs.

4.6 Test methods and procedures.

4.6.1 Test conditions. Unless otherwise specified herein, the following conditions shall be used as a basis to establish performance requirements:

a. Temperature, room ambient plus or minus 9 degrees Celsium (C) (plus or minus 16 degrees Fahrenheit (F)).

b. Altitude, facility ground.

c. Humidity, facility ambient up to 95 percent relative humidity.

4.6.2 Test sequence. Testing within each classification of inspection (preproduction or acceptance) shall be at the option of the supplier.

4.6.3 Tests.

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4.6.3.1 Composition and condition in container. The composition shall be certified by the supplier and condition in the container shall be determined visually to establish compliance with 3.2.1.

4.6.3.2 Specific gravity. The specific gravity of the compound shall be determined in accordance with ASTM D 70 and shall conform to 3.2.2.

4.6.3.3 Thermal conductivity. The thermal conductivity of the material shall be determined using the following test equipment and procedures.

4.6.3.3.1 Equipment.

a. Conductivity cell in accordance with Figure 1.

b. Voltmeter 0-1.5 volts.

c. Ammeter 0-5 amperes.

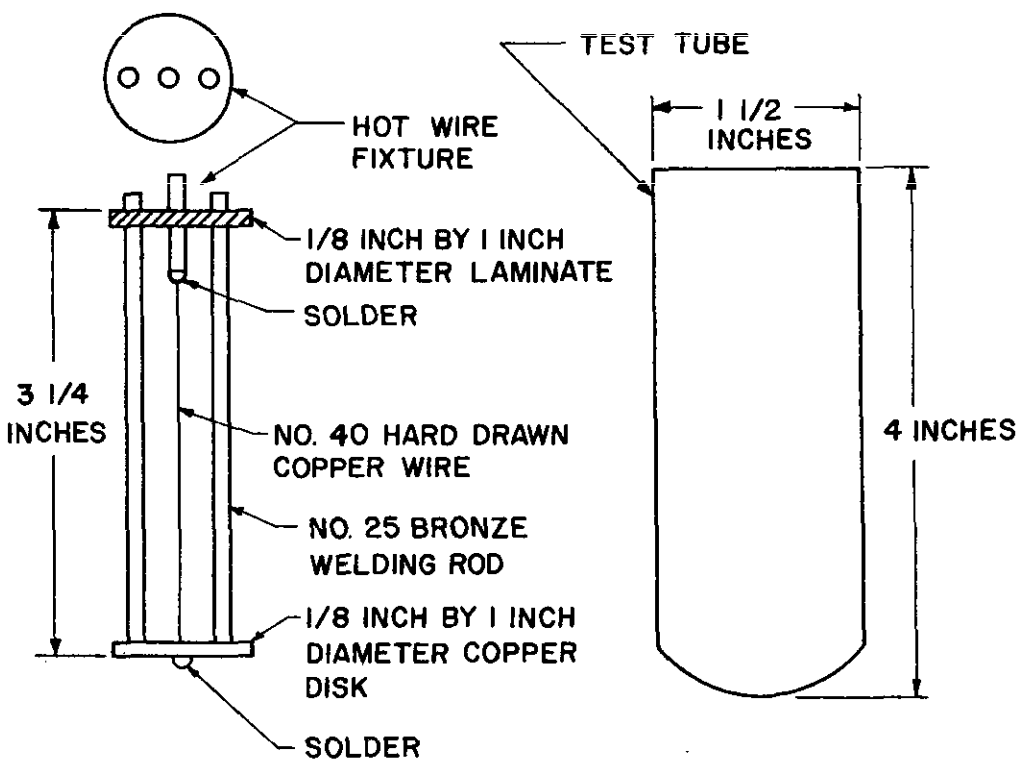


Figure 1. Thermal conductivity test - hot wire method.

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4.6.3.3.2 Procedure.

a. A sample tube shall be filled with the compound to within ¾ inch of the top.

b. The hot wire fixture shall be inserted into the tube until the compound is at the same level as the top insulating disc.

c. The tube containing the compound and fixture shall be placed into a 400-millimeter (ml) beaker containing water maintained at 23 degrees plus or minus 1 degree C (73 degrees plus or minus 2 degrees F).

d. When the tube and the compound reach 23 degrees plus or minus 1 degree C (73 degrees or minus 2 degrees F), the voltage shall be raised to 1.3 volts provided current flow does not exceed 4 amperes.

e. The voltage shall be allowed to continue until current is stabilized. At this point voltage and current readings shall be recorded employing diagram in Figure 2.

f. Voltage shall be reduced by 0.1 volt and after 2 minutes current flow shall be recorded.

g. Voltage reduction shall be continued in 0.1-volt increments until 0.5 volt is reached. Voltage and current readings shall be recorded after each reduction as specified in step (f).

h. R_C, T and W shall be calculated, and T versus W shall be plotted to determine slope in accordance with the following calculations:

4.6.3.3.3 Calculations.

$$R_C = \frac{E}{I} \qquad W = EI$$

$$T = \frac{R_C \times 257}{R_E} - 234.5$$

R_C = circuit resistance = $\frac{E}{I}$

R_E = resistance of wire at 23 degrees C (73 degrees F). This resistance shall be measured by means of a bridge. Resistance of test leads shall be subtracted when determining resistance of wire.

W = Watts

E = Voltage (volts)

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I = Current (amperes)

T = Temperature in centigrade

Plot temperature versus watts and calculate $\frac{W}{T}$

$\frac{W}{T}$ = Slope

Thermal conductivity $K = \frac{0.00038}{0.022} \times \text{Slope}$

Units of K are : Calories - cm
Second .cm^2 °C

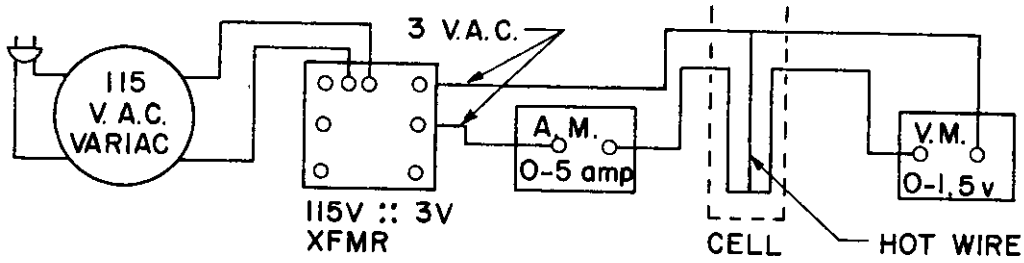


Figure 2. Thermal conductivity test diagram

4.6.3.4 Color. The color shall be determined visually to establish compliance with 3.2.4.

4.6.3.5 Penetration. Penetration of the compound shall be determined in accordance with ASTM D 217. The test shall be run 1 minute after working, and results shall conform to 3.3.1.

4.6.3.6 Bleed. Bleed of the compound shall be determined in accordance with Method 321 of Federal Test Method Standard No. 791, with the following exceptions:

a. The cone shall be suspended from a rod supported on the edges of the beaker, without covering the beaker.

b. The test shall be conducted at 392 degrees plus or minus 1 degree (200 degrees plus or minus 0.5 degrees C) for at least 24 hours.

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$$\frac{\text{Gain in weight of beaker}}{\text{Weight in sample}} \times 100 = \%\ \text{Bleed}$$

Bleeding of the compound shall conform to 3.4.1.

4.6.3.7 Evaporation. Evaporation of the compound shall be determined via the procedure given in paragraph 4.6.3.6.

$$\text{Percent Evaporation} = \frac{W_1 - W_2}{W_3} \times 100$$

W_1 = Wgt. of beaker, cone assembly plus compound before heating

W_2 = Wgt. of beaker, cone assembly plus compound after heating

W_3 = Wgt. of beaker, cone assembly plus compound minus the wgt. of the beaker, cone assembly = Wgt. of samples

4.6.4 Workmanship. The compound shall be visually examined for conformance to workmanship requirements as specified in 3.5.

4.7 Packaging, packing and marking. The inspector shall ascertain that the packaging, packing and marking of the material conforms to this specification.

5. Packaging.

5.1 Preservation, packing, unitization and marking. Level A or B (see 6.2).

5.1.1 Level A or B. Preservation, packing, unitization and marking shall be in accordance with VV-D-1078 and specifications referenced therein.

6. NOTES

*6.1 Intended use. This material is intended for use in improving thermal conductivity at heat sink junctions. Caution: When this material is initially applied under devices which are torque set, the zinc oxide filler may not allow complete seating of the device. The following procedure for use should be adhered to:

Apply a thin film of silicone and/or non-silicone grease, then torque to the pressure specified on the device detail drawing; after 20 - 30 minutes, retorque to specified pressure.

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*6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Whether a first article sample is required (see 3.1 and 4.3), and if so, pertinent details.

c. Requirement for certification of composition (see 4.6.3.1).

d. Type and site of container desired.

e. Level of packaging required (see 5.1).

6.3 Changes to previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modification, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-47113B(MI)	Compound, Heat Sink, Silicone and/or Non-Silicone

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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